

Evolving with Generative AI*

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Abstract

How do humans adapt their behavior in response to AI upgrades, and how does this adaptation affect creative output? I study this question using prompt-level data from Midjourney, a leading AI image generator, where users interact with AI by submitting combinations of words known as prompts. Textual analysis shows that users systematically change how they write prompts following AI upgrades. Consider the image production function as $Image = F(AI, Prompt)$. I ask how much of the change in output is attributable to ΔAI versus $\Delta Prompt$. By submitting prompts written for the old AI to the new AI and vice versa, I decompose the output shifts as arising from prompt changes (73%), AI changes (20%), and a residual (7%), implying complementarity between AI and human inputs. The decomposition should be interpreted as a first-order expansion of the production function. More generally, this paper provides a new way to understand the production function without explicitly assuming a functional form.

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1 Introduction

In recent years, generative AI has advanced rapidly, especially in language-based models. Between November 2022 and October 2025, five major versions of GPT have been publicly released. Language-based AI image, video, and music generators are also developing rapidly, with each new version demonstrating greater capability than its predecessor. Generative AI is now being used for creative tasks, including creative writing, email drafting, and artistic production. These were once considered tasks that could only be completed by humans. Creative industries have increasingly adopted generative AI into their workflows. For instance, AI companies are partnering with studios in film, gaming, and news to create illustrations and animated characters. With AI being widely adopted and improving so quickly, how do humans adapt to AI upgrades? How does such adaptation impact the output of AI-assisted production?

In this paper, I study user behavior on Midjourney, a leading AI image generator. Midjourney operates in a public chat channel on Discord, a messaging platform. In this environment, users submit prompts, which are a combination of words, in the chat box to the AI, which then returns generated images in exchange for a fee. This setting enables me to observe the submitted prompts and generated images. The richness of this data allows me to study how humans interact with generative AI. During my observation periods, four new versions of Midjourney were released. In particular, the transition from version 3 to version 4 represented a large improvement in AI capability. I measure this improvement by comparing how closely generated images align with the prompts that produced them. The changes across Midjourney versions enable me to investigate how users modify their prompts in response to AI upgrades. The setting also allows me to resubmit prompts written for version 3 to version 4 and prompts written for version 4 to version 3. These resubmissions allow me to observe counterfactual outputs without imposing model assumptions.

When AI upgrades, its capabilities improve, and the generated images change for a given prompt. The words that users include in their prompts also change systematically after AI upgrade. For example, users include more words about colors

and photography in the new AI than in the old AI, such as “neon color” and “shutter speed”. This is because the new AI understands these words better. Conversely, users include fewer words like “detailed” and “realistic” because the new AI automatically makes images more detailed and realistic without users mentioning these words in the prompt.

Consider the creative process as a production function, where the inputs are AI capability and human input, and the output is the generated image. Both AI capability and human input change simultaneously in the new AI, jointly changing the generated images. This raises the research question: how much of the image change is due to the change of AI and how much is due to the change of human input?

To decompose these changes, I resubmit prompts written for the old AI to the new version and vice versa, collecting the resulting images, which are then converted into embedding vectors using large language models. I project the vectors onto the direction of $\overrightarrow{Image_{new}} - \overrightarrow{Image_{old}}$, which captures the overall shift in image output between AI versions.

This decomposition reveals that 73% of the changes in images result as a consequence of changing prompts alone, 20% from changing AI alone, and 7% from residual effects. These findings demonstrate that human input remains highly relevant even as AI capabilities advance. Simply applying historical prompts to new AI cannot replicate the current outputs. The positive signs of these contributions indicate that AI and human input are complements rather than substitutes. When AI learns to understand more words, users begin using those words more often to take advantage of its expanded vocabulary. The AI upgrade from version 3 to version 4 increases the return of human input.

More generally, this paper provides a framework for modeling the production process without assuming a functional form. This paper asks whether we can learn about the production process without imposing a functional form, by directly observing the two counterfactuals $F(k_{new}, l_{old})$ and $F(k_{old}, l_{new})$. The decomposition can be interpreted as an approximation to the first-order expansion of the production function. This approach relates to the literature on production function estimation (Acker-

berg, Caves, & Frazer, 2015; Hsieh & Klenow, 2009; Levinsohn & Petrin, 2003; Olley & Pakes, 1996; Wooldridge, 2009). This literature typically estimates production functions by imposing a functional form and addressing the endogeneity of observed input choices, such as capital and labor, to unobserved productivity. In contrast, I use experimentally generated counterfactual outputs to study the production process without specifying a functional form.

In summary, this paper studies how humans adapt to new AI technology and its consequences on production. By submitting old prompts to new AI and new prompts to old AI, I decompose the output shifts and show that human inputs and Midjourney are more like complements than substitutes. The results suggest that human judgment plays a significant role as technology improves. When AI upgrades, humans are still needed to judge what the right words are to submit to AI.

Related Literature

This paper connects to three strands of literature. First, I contribute to recent work examining usage patterns in generative AI (Chatterji et al., 2025; Handa et al., 2025). This literature analyzes conversational data between users and AI to understand the purpose of usage. Handa et al. (2025) document that AI usage concentrates primarily in software development and writing tasks. Chatterji et al., 2025 find that users mostly employ generative AI to seek information and refine writing.

While I also analyze conversational data, my focus differs. Rather than examining why users employ AI, I investigate how users interact with AI. In Section 5, I show that users adapt to generative AI by systematically modifying the words in their prompts. When AI upgrades, users adjust their prompts to better leverage the new capabilities.

Lin (2026) uses conversational data from Midjourney to study how users iteratively adjust their prompts *within* a production unit to converge toward a desired outcome. The paper finds that human learning in the loop speeds up the production process by three times. While this paper also uses conversational data from Midjourney, its focus is different. Rather than examining prompt adjustments *within* a

session, I study how users change their prompts across AI versions and how these changes contribute to shifts in output. The focus is therefore on how human labor adapts in response to technological change.

Second, I contribute to the literature exploring AI’s impact on productivity and labor markets. This body of work primarily evaluates AI through three lenses: disruption of traditional labor (Acemoglu and Restrepo, 2020; Acemoglu, Autor, Hazell, and Restrepo, 2022; Babina, Fedyk, He, and Hodson, 2024), enhancement of professional output (Agrawal, Gans, and Goldfarb, 2019; Peng, Kalliamvakou, Cihon, and Demirer, 2023; Brynjolfsson, Li, and Raymond, 2025), and experimental task-based performance (Dell’Acqua et al., 2023; Noy and Zhang, 2023; Kanazawa, Kawaguchi, Shigeoka, and Watanabe, 2025). Additionally, several studies examine the nuances of human-AI collaboration in specialized domains like recruitment and legal services (Angelova, Dobbie, and Yang, 2023; Vaccaro, Almaatouq, and Malone, 2024; Choi and Schwarcz, 2024).

This literature relates to the decomposition described in Section 6, where the production function of AI-generated images is modeled as a function of both AI and human input. The decomposition aims to isolate the contribution of human input within this production process. Existing studies (Acemoglu and Restrepo, 2020; Acemoglu et al., 2022; Babina et al., 2024; Dell’Acqua et al., 2023; Noy and Zhang, 2023; Brynjolfsson et al., 2025) generally compare outcomes with and without AI adoption, typically holding AI capability constant. These analyses focus on predetermined tasks and often overlook the possibility that users pursue increasingly complex goals as AI capabilities advance. Rather than comparing outcomes with and without AI, I focus on how users adapt their behavior across AI versions, conditional on AI adoption.

The study most closely related to my decomposition analysis is Jahani et al. (2025), which implements an experiment collecting user prompts from an older AI version and reusing them in a newer one. In their setting, task goals are fixed by the researchers, and image quality is evaluated by the similarity between generated and target images. By contrast, while I also resubmit old prompts to newer models and vice versa, since this is observational data instead of an experiment with a fixed task,

the goals users pursue can change with technology. As AI improves, users aim for more complex goals that were previously infeasible. For instance, Midjourney V3 did not recognize photographic terms such as “10mm lens,” so users did not include them to control the angle of image views. However, Midjourney V4 can interpret these terms, and users actively include them to manipulate viewing angles more precisely, as shown in Section 5. Both AI capabilities and human inputs evolve when the AI upgrades. Users change their prompts to take advantage of the new technology and achieve more complex goals. In Section 6, I decompose their relative contributions to creative outcomes and show that, when AI upgrades, over 70% of the image change is due to users changing what they write in the prompts, rather than to the AI being different.

Third, this paper contributes to the literature of production function estimation (Olley and Pakes, 1996; Levinsohn and Petrin, 2003; Hsieh and Klenow, 2009; Wooldridge, 2009; Akerberg et al., 2015). This literature develops methods to estimate a production function with an assumed functional form. A key insight is that observed choice variables, including capital and labor, are endogenous to unobserved productivity. This literature raises this challenge and brings different solutions.

While this paper is also about production functions, I do not explicitly assume a functional form. Instead, I try to understand the production function by utilizing the two counterfactuals $F(k_{new}, l_{old})$ and $F(k_{old}, l_{new})$, which can be exogenously obtained by collecting previously submitted prompts and resubmitting them into different AI versions in the setting of Midjourney.

The remainder of the paper is structured as follows. I introduce the empirical setting and data in Section 2 and Section 3. Section 4 verifies that newer AI gives images closer to the submitted prompts in terms of meaning. Section 5 presents evidence that users change words in the submitted prompts when AI upgrades. Section 6 decomposes output shifts to quantify how much of the image difference is driven by the change of AI and how much by the change of prompts. Section 7 concludes.

2 Empirical Setting

The context studied in this paper is Midjourney, one of the leading AI text-to-image generators, which operates on a messaging platform. In this context, both the prompts submitted by users and the corresponding image outputs are observable.

Midjourney allows users to submit prompts, typically text and sometimes image references, to generate images using its AI model in exchange for a fee. Prior to August 2024, Midjourney was only accessible on a public chat channel on Discord, a messaging platform that typically allows game streamers to communicate with each other in real-time. After Midjourney opened its own web interface to all users in August 2024, a significant number of users have continued to use Discord to access Midjourney’s services. There are two major reasons for the partnership between Midjourney and Discord. Anecdotally, during its beta phase, Midjourney operated as a small firm with limited financial support. Operating with Discord’s infrastructure can be cost-saving. Besides, according to an interview of the company’s founder and CEO, David Holz, a key motivation was to build an interactive community where users could discuss ideas and prompt crafting techniques.¹

The public chat channel allows any user to join, and users see each other’s actions. It enables researchers to observe the creative process of art generation using AI. Figure D2 in the appendix shows an example of Midjourney usage. When a user submits a prompt through the chat box, the AI returns four generated images within minutes, along with some interactive buttons. For each prompt and each button clicked, I observe a timestamp and a corresponding user ID.

This empirical setting offers several benefits for research. First, it allows direct observation of inputs and outputs of creative production, including historically submitted prompts and generated images.

Second, during the observation period, four new versions of Midjourney were released. These changes enable me to investigate how users modify their prompts in response to AI updates. In the literature, researchers have been utilizing the

¹<https://www.theverge.com/2022/8/2/23287173/ai-image-generation-art-midjourney-multiverse-interview-david-holz>

variation in production with and without AI to estimate the effects of AI adoption (Angelova et al., 2023; Dell’Acqua et al., 2023; Zhou and Lee, 2023; Noy and Zhang, 2023; Peng et al., 2023; Babina et al., 2024; Choi and Schwarcz, 2024; Vaccaro et al., 2024; Brynjolfsson et al., 2025; Kanazawa et al., 2025; Zhou, Lee, and Gu, 2025). These papers generally do not capture how users adapt their behaviors as AI evolves. This setting allows me to understand how users dynamically adjust their behaviors in response to AI version changes in Section 5, conditional on already using the technology. When AI upgrades, users systematically include different words in their prompts. They adapt their prompts to take advantage of the new technology.

Third, this setting allows researchers to observe two counterfactuals directly without imposing model assumptions. In conventional settings, researchers can observe outputs from old machines, old labor input $F(K_{old}, L_{old})$; and the outputs after firms adopt new technology and labor is adjusted based on new technology $F(K_{new}, L_{new})$. By contrast, this setting allows me to observe outputs from new machines with old labor input $F(K_{new}, L_{old})$, and old machines with new labor input $F(K_{old}, L_{new})$. This is achieved by submitting the old prompts to the new Midjourney version and submitting new prompts to the old Midjourney version. By observing these counterfactuals, I am able to decompose output changes into effects from prompt changes and effects from AI changes in Section 6. When AI upgrades, the generated images change. The decomposition shows that most of the image differences come from users changing their prompts rather than from changes in AI itself. The decomposition result implies complementarity between AI and human input.

Finally, Midjourney is both influential and representative. Although there is no official information about Midjourney’s market share, Midjourney has been one of the “most popular AI image generators” upon search results, along with DALL-E, Stable Diffusion, Adobe Firefly, and Leonardo AI.² By October 2025, Midjourney has become the largest Discord server by number of users.³

It is useful to understand who uses Midjourney and the purpose of using it. Al-

²<https://www.demandsage.com/ai-image-generators/>; <https://www.cmswire.com/digital-marketing/a-marketers-guide-to-the-best-text-to-image-generators/>; <https://www.forbes.com/lists/ai50/>

³<https://backlinko.com/discord-users>; <https://influencemarketinghub.com/discord-stats/>

though Midjourney does not have an official report about the composition of users, there are “micro-polls” and a “discussion” sub-channel in the Midjourney Discord server.⁴ There are also some online discussions on social media like Reddit, Twitter, and Instagram. Additionally, SimilarWeb, a data analytics company, provides information on what other websites Midjourney users frequently visit. Together, these sources help us understand the user composition. Some screenshots of the “micro-polls” and the SimilarWeb report can be found in Appendix C.

Many Midjourney users possess formal art skills. They self-reveal that they are currently working in or used to work in the art or design industries. They own professional drawing tablets like Wacom and maintain Adobe Photoshop subscriptions. Apart from Midjourney, they also frequently browse other AI tools like ChatGPT, graphic design platforms, and art resource websites such as Behance, e-commerce platforms for selling custom products like T-shirts and mugs, and freelancer websites like Fiverr. Many of these users are hobbyists and professional users. There are some other types of creators, such as fiction writers and musicians. There are also small business owners.

Users engage with Midjourney for various purposes. Many of them are using Midjourney purely for fun. Some users are using Midjourney to create decorative images for social media posts, wallpapers, and T-shirts. Sometimes they sell these creations. Some users use Midjourney to accelerate the sketching process. They experiment with combinations of colors and compositional elements before transferring the concepts to physical canvases. Some creations are for commercial purposes, such as book illustrations, music album covers, and logo designs. In some cases, users employ Midjourney as a source of creative inspiration. For example, fiction writers use it to visualize scenery, and musicians generate visual concepts to inform their musical compositions.

One important point to note is that Midjourney’s official documentation from version 3 through version 6 contains no reference to session-based memory. In particular,

⁴These polls are usually initiated by Midjourney’s CEO David Holz. According to the description of this sub-channel, Midjourney is using it to construct larger questionnaires, which he later publishes in the “announcements” sub-channel for users to fill out.

the company does not explicitly state that a user’s previous prompts influence outputs generated in subsequent interactions. Midjourney does not implement memory in the same way as session-based systems like ChatGPT. To modify a prior result, users must manually reconstruct context. This can be done either by copying and pasting a previous output as part of a new prompt or by copying and editing the original prompt to create a new, independent query. In both cases, each prompt is treated as an independent input without persistent session memory. Personalization features were not introduced until version 7.

3 Data

The data includes all publicly available messages from the Midjourney server on Discord, between June 2022 and January 2025. Each message sent by the Midjourney Bot represents an action of a user. An action could be submitting a prompt to Midjourney or clicking a button below the generated images. For each action, I can observe the timestamp, user ID, submitted prompt, generated images, and the buttons clicked, if any. In this paper, I primarily focus on prompts submitted to Midjourney rather than button interactions.

One caveat is that users who purchase higher-tier subscriptions may activate “stealth mode”, which conceals their actions from the public chat channel. These messages are not included in the data. A natural concern is whether users worry about others observing and appropriating their prompts. Discussions on platforms such as Reddit suggest that a high volume of activity in the channel, where individual messages are quickly displaced by hundreds of others, mitigates such concern. Users generally do not appear to think of prompt imitation as a significant risk.

In the remainder of the paper, I focus on a random sample of 500 users who submitted at least 50 prompts in each of V3, V4, V5, and V6 in the dataset, along with their complete usage history. There are three reasons for this sampling strategy. First, it minimizes the impact of users who tried Midjourney briefly out of curiosity before quitting, allowing me to focus on users with sustained engagement. Second, it ensures every user has sufficient usage in each Midjourney version, such that I can

analyze their behavioral changes as the AI improves. Third, given the computational intensity of processing text and image data, restricting the sample to a small subset makes subsequent data processing feasible.

An illustrative example of a prompt is as follows.

```
**<https://s.mj.run/ABCDEFGF> dark fantasy, retro anime style, mink  
hiding in a forest, horror core, ultra realistic, retro, video game style, creepy -  
- aspect 5:4** - <@123456789987654321>
```

07/16/2023 11:18:25.796 UTC

In this example, “https://s.mj.run/ABCDEFGF” is the link to the image input; “dark fantasy, retro anime style, mink hiding in a forest, horror core, ultra realistic, retro, video game style, creepy” is the text input; “aspect 5:4” is a parameter, and “123456789987654321” is the user ID. Note that there can be multiple image inputs and multiple parameters in a prompt. Every prompt is associated with a timestamp. In the following analysis, I focus on the text input of the prompts.

3.1 Focusing on the Final Prompt in a Session

One feature of prompt writing is that users seem to adjust the prompts iteratively to converge to their desired outcome. One example could be the following.

1. “a closeup of a book cover on a beach on a rainy day”
2. “closeup of a book cover lying in the sand on a beach”
3. “closeup of a book cover lying in the sand on a beach, an hourglass is also in the sand”
4. “closeup of a book cover lying in the sand on a beach, a small hourglass is also in the sand”

This behavior is consistent with online discussion. Users adjust words in a prompt or rerun prompts until the output better matches their desired outcome. Hence, I treat a sequence of correlated prompts submitted by one user within a short period of time as a production unit called a session. The prompts within a session are intermediate steps of production.

I identify prompt sessions by computing Jaccard similarity across each user’s prompts and clustering them using DBSCAN. The intuition is that a sequence of similar prompts from the same user represents a single production unit. I validate the resulting sessions using timestamps: over 90% of the sessions are completed within 24 hours. The details of session construction can be found in Appendix A.3. Table B1 in the appendix represents the summary statistics.

While the prompt evolution *within* a session is also important, it is not the focus of this paper. Instead, I study how the *final* outcome of a production unit changes following an AI upgrade. Accordingly, the main analysis in Section 6 uses only the final prompt in each session. A more detailed analysis of within-session prompt evolution is provided in Lin (2026).

3.2 Supplementary Data

Apart from messages on the Midjourney Discord server, I also use supplementary data from online Midjourney community classifications, including Midlibrary, Imiprompt, ShellyPalmer, and a popular collection of words on a GitHub repository.⁵ These data give me information like artist names, photography words, and style words, which are later used in prompt analysis.

⁵<https://midlibrary.io/>;
<https://www.imiprompt.com/>;
<https://shellypalmer.com/midjourney-reference-art/>;
<https://github.com/willwulfken/MidJourney-Styles-and-Keywords-Reference/blob/main/README.md>

4 Newer AI is More Accurate

AI models are developing rapidly, and more recent models exhibit greater capabilities. During the observation period, Midjourney released six versions: V1, V2, V3, V4, V5, and V6. Higher version numbers indicate more recent and capable models, which is similar to the difference between GPT-4 and GPT-5. V1 and V2 were primarily used during the invite-only closed beta test on Discord. On July 12, Midjourney entered open beta, allowing anyone to join the Discord server. Shortly after the beginning of open beta, Midjourney released V3 on July 25, 2022. Since Midjourney did not disclose the selection criteria for the closed beta test, this paper focuses on user behavior from V3 through V6. A user can freely choose any version that has become available by the time of usage, which means that she has access to previous versions. The data allows me to observe which version the users employ. Only 1.6% of the prompts are using the version below the latest default one. By revealed preference, users seem to prefer the later versions.⁶

Figure 1 illustrates differences across the six versions. The same prompt “Satyr, ancient greek tavern, dickensian fashion, cosmic horror, non-euclidian geometry” was submitted to each version, generating substantially different images. More recent versions tend to give more detailed images that appear to align more closely with the prompt. There is a significant improvement between V3 and V4. Other than that, there is improvement between V4 and V5, V5 and V6 as well.

To quantify this improvement in AI capabilities, I employ text and image embeddings, a standard and widely adopted approach in text and image analysis. The method converts text or images into vectors in a high-dimensional space using large language models or vision-language models. The embeddings, which are vectors, are

⁶V3 was released on July 25, 2022, and became the default version immediately. V4 was released on November 5, 2022, and became the default version on December 20, 2022. V5 was released on March 15, 2023, and became the default version on May 4, 2023. V6 was released on December 21, 2023, and became the default version on February 14, 2024.

I infer which version a user employed based on timestamps and prompt parameters. For example, if a prompt contains “- V5” the user is employing V5 even if the timestamp falls after February 14, 2024. If a prompt does not specify a version, it uses the default version for that time period. Figure D1 in the appendix displays the version timeline.



(a) V1



(b) V2



(c) V3



(d) V4



(e) V5



(f) V6

Figure 1: Illustration of Version Differences

Notes: This figure presents image outputs from Midjourney with the **same prompt** in **different versions**: “Satyr, ancient greek tavern, dickensian fashion, cosmic horror, non-euclidian geometry”.

interpretable through cosine similarity between them. If two pieces of text have high cosine similarity, they are similar in meaning space, as illustrated in Figure 2. In this paper, I use OpenAI text-embedding-3-small for the main results. In the appendix, I use all-MiniLM-L6-v2, another embedding model, to cross-verify robustness. Results are robust to both embedding models.

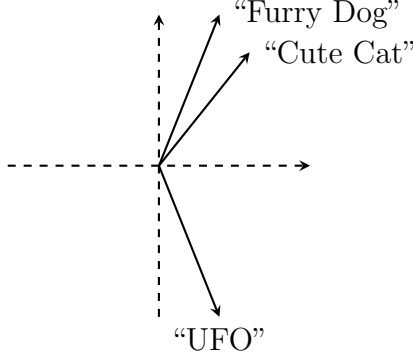


Figure 2: Embedding Illustration

The following experiment is implemented. I randomly select 500 prompts from the data and submit each prompt to all six MidJourney versions. I then collect the generated images and ask GPT-4o to “describe the image”. This approach enables me to compare the input (prompt) and output (proxied by the GPT-generated description) and quantify how AI capability changes across versions using regression (1).

$$\text{Cosine Similarity}(\vec{p}_{\text{prompt}}, \vec{p}_{\text{GPTdescription}}) = \sum_v \beta_v \mathbf{1}(\text{Midjourney version} = v) + \mu_{\text{prompt}} + \epsilon_{pv} \quad (1)$$

In regression (1), the dependent variable is the cosine similarity between the submitted prompt and the GPT-generated description of the resulting image. Cosine similarity, which takes values between -1 and 1, is computed from text embeddings generated by OpenAI’s text-embedding-3-small model. Values closer to 1 indicate

that the two texts are similar in meaning, whereas values closer to -1 denote opposite meanings. Higher cosine similarity indicates greater accuracy: Midjourney produces images that align more closely with the input prompt. The independent variables are dummies indicating Midjourney versions, along with the prompt fixed effects.

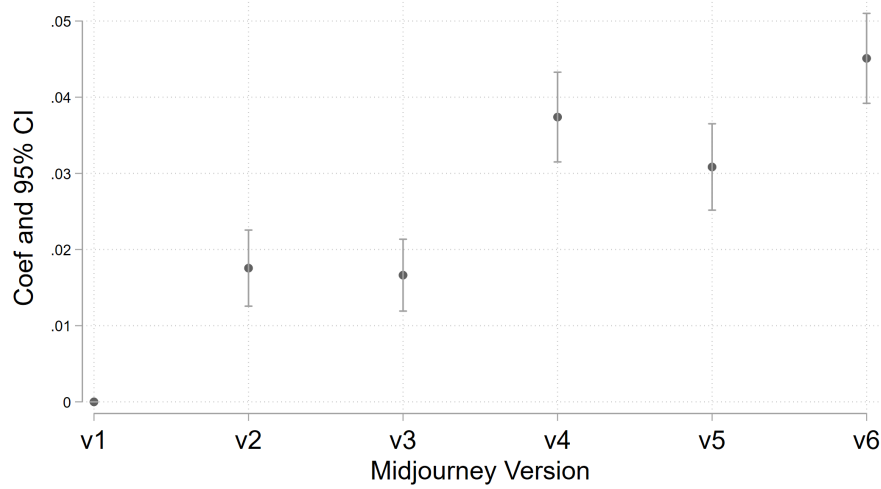


Figure 3: Later Versions Give More Accurate Results

Notes: This figure plots coefficients before 1(*Midjourney version* = *v*) in regression (1) and the 95% confidence intervals. Standard errors are clustered at the prompt level. This figure uses embeddings from the embedding model text-embedding-3-small from OpenAI. The dependent variable, cosine similarity, can range from -1 to 1. Values closer to 1 indicate that the two texts are similar in meaning, whereas values closer to -1 denote opposite meanings. Higher cosine similarity indicates greater accuracy: Midjourney produces images that better align with the input prompt. I run the same regression with embeddings from all-MiniLM-L6-v2. The result is reported in Figure B1 in the appendix.

Figure 3 plots the coefficients of β in regression (1). More recent Midjourney versions generally produce more accurate images corresponding to the submitted prompts. As expected, there is a substantial increase in accuracy between V3 and V4.

A more recent Midjourney version generally implies a more accurate image corresponding to the submitted prompt. As expected, there is a big jump between V3 and V4. A potential mechanism is that more recent versions generate more detailed

images. In Figure B2 in the appendix, I plot the similar coefficients in regression (1), with the dependent variable being the length of the GPT description. The results suggest that more recent versions tend to generate images requiring more words to describe, hence greater detail. Although a more detailed image does not always imply that it is more aesthetic, a more accurate output at least represents a more faithful reflection of the submitted prompt.

In Section 5 and Section 6, I exploit this substantial shift between V3 and V4. I show that users change prompts, and that these changes in AI capability and prompt wording jointly and systematically alter the generated images. By submitting prompts written for the old AI to the new version and vice versa, I decompose shifts in output images into contributions from AI improvements and prompt adaptations.

5 People Change Words that They Use to Prompt in New AI

Since different Midjourney versions have different capabilities and recognize different vocabularies, users write significantly different prompts across versions. Users of the online Midjourney community have voluntarily summarized the vocabulary each version recognizes. For example, websites document how each style appears across different Midjourney versions.⁷ There is also a public Google spreadsheet summarizing whether specific artist names are recognized in each version.⁸ Online discussions also debate which “magic words” enhance artistic quality in each version. The existence of these community resources suggests that users actively explore the effect of putting various words in the prompts across Midjourney versions.

To empirically show that words in the prompts are systematically different across Midjourney V3 and V4, I construct word classifications using the Midjourney community online resources, including Midlibrary, Imiprompt, ShellyPalmer, and a popular collection of words on a GitHub repository summarized by Will Wulfken. For each

⁷<https://midlibrary.io/>

⁸<https://docs.google.com/spreadsheets/d/1cm6239gw1XvvDMRtazV6txa9pnejpKkM5z24wRhhFz0>

user, I consider the 50 prompts before and the 50 prompts after the V4 release date, constructing a balanced panel around this threshold. Using these classifications and the balanced panel, I start by estimating a sample regression 2 with the dependent variable being one of the most popular types of words in Midjourney V3: “realistic words.”

$$\#Realistic\ Words_{ip} = \sum_k \beta_k \mathbb{1}(k^{th}\ Prompt)_{ip} + \mu_i + \epsilon_{ip} \quad (2)$$

The dependent variable is the number of “realistic words” in prompt p written by user i . Realistic words include terms such as “realistic”, “super-realistic”, “hyper-realistic”, etc.⁹ The independent variable $\mathbb{1}(k^{th}\ Prompt)_{ip}$ is a set of dummy variables indicating whether this is the k^{th} prompt since the V4 release date of user i . The index k can be negative if the prompt is written before the V4 release date. The μ_i are the user fixed effects. Panel (a) in Figure 4 plots the estimated coefficients β_k against k .

The result shows that users systematically include fewer realistic words in prompts for V4 than V3. This pattern likely reflects the fact that V4 generates more realistic images than V3 by default, reducing the need to explicitly specify “realistic” in prompts.

Similar patterns are found in other dimensions. Figure 4 shows that users include fewer terms such as “detailed” or “octane render” in prompts for V4, since the images produced by V4 are more detailed by default. Conversely, users employ more color and photography-related terms in V4 prompts because V4 interprets these words more effectively. Users also include more stop words, such as “a” “the” “here” “there”, in their prompts. This is because V4 understands natural language better, and users are doing less prompt engineering. For example, a V3 prompt might be crafted into “cat, blue shirt, sitting at table, white background”. Whereas a V4 prompt can be phrased as “a cat wearing a blue shirt sitting at a table with a white background”. The increased presence of stop words suggests that users are crafting prompts into sentences that resemble natural language, as the new AI understands

⁹The list can be found in Appendix A.5

them more effectively. This systematic change in the words of the prompt suggests that users are adapting to new AI by adjusting their actions.

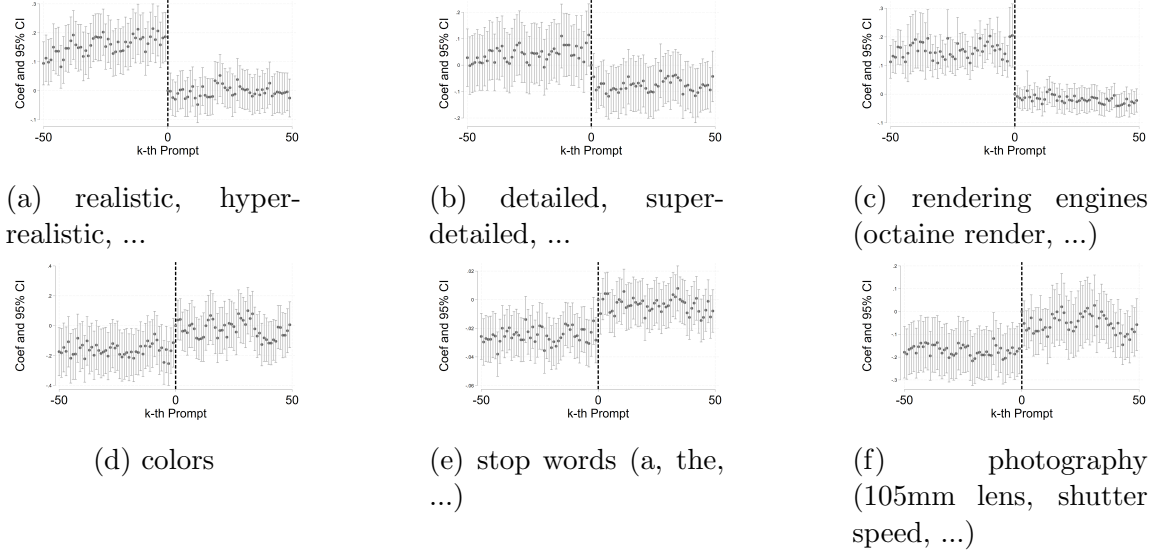


Figure 4: People Change Prompts in New AI

Notes: The classifications of words can be found in Appendix A.4.

Several mechanisms may explain why users adjust their behavior following a new release. When the new version is released, Midjourney usually publishes an updated user manual, highlighting the new features. Users may read this manual and modify their behavior accordingly. In addition, many users write blogs or post YouTube videos comparing different Midjourney versions, which can serve as learning resources. Furthermore, given the public nature of the Discord chat channel, users may also observe other users' prompts and outputs and learn from each other.

This chapter abstracts away from discussing these underlying channels. Instead, it takes behavioral change as given and focuses on how such changes affect creative outcomes. The discussion of the channels remains for future research. Throughout this chapter, any change in user behavior in response to a technology change, regardless of its underlying channels, is referred to as adaptation.

6 Decomposition of Output Shifts

When AI upgrades, both AI capability and words in the prompt change, jointly altering the generated images. This section decomposes total shifts in output images into contributions from AI changes alone and contributions from prompt changes alone. A key feature of this setting is that, unlike typical settings where researchers observe only outputs from old machines and old labor input ($Y = F(K_{old}, L_{old})$) and outputs from new machines and new labor input ($Y = F(K_{new}, L_{new})$), I can directly observe two counterfactuals without imposing model assumptions: outputs from new machines and old labor input ($Y = F(K_{new}, L_{old})$), and outputs from old machines and new labor input ($Y = F(K_{old}, L_{new})$). To construct these counterfactuals, I submit prompts users wrote for the old AI to the new version, and vice versa.

Specifically, I collect images users generated with the old AI, $F(AI_0, L_0)$, and images generated with the new AI $F(AI_1, L_1)$. I then submit the old prompts to the new AI to obtain $F(AI_1, L_0)$, and submit new prompts to the old AI to obtain $F(AI_0, L_1)$.

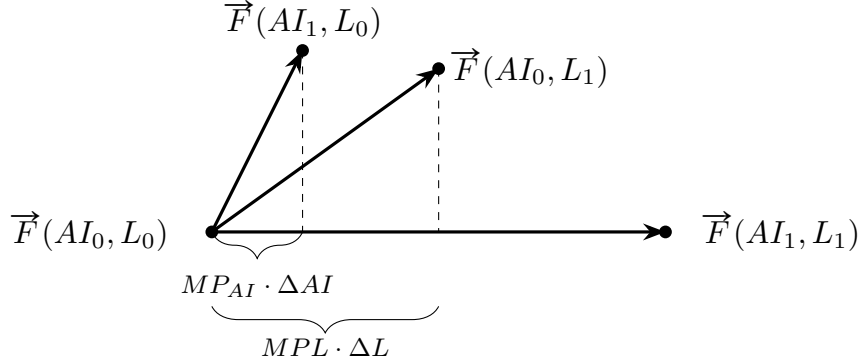


Figure 5: Projection Illustration

The collected images are first described using GPT-4o, and these descriptions are then converted into vector embeddings with OpenAI’s text-embedding-3-small model. The vector embeddings are representations in meaning space. The vectors are then projected onto the direction of $\vec{F}(AI_1, L_1) - \vec{F}(AI_0, L_0)$ as illustrated with Figure 5. Define the vector from old output $\vec{F}(AI_0, L_0)$ to the new output $\vec{F}(AI_1, L_1)$

as representing 100% of the output shift, $MP_{AI} \cdot \Delta AI$ captures the output shift attributable to AI changes alone, marginal product of AI multiplied by AI changes. While $MPL \cdot \Delta L$ captures the output shift attributable to prompt changes alone, the marginal product of labor multiplied by labor changes. The next section provides an economic interpretation of this decomposition.

6.1 Decomposition: Interpretations

This decomposition can be interpreted as a discrete approximation of the first-order expansion of the production function, as shown in equation (3). When Y represents output level, $MP_{AI} \cdot \Delta AI$ corresponds to the marginal product of AI multiplied by the change in AI. $MPL \cdot \Delta L$ corresponds to the marginal product of prompts multiplied by the change in prompts. ϵ captures the residual.

$$Y_1 - Y_0 = \underbrace{\frac{\partial F(AI_0, L_0)}{\partial AI}(AI_1 - AI_0)}_{MP_{AI} \cdot \Delta AI} + \underbrace{\frac{\partial F(AI_0, L_0)}{\partial L}(L_1 - L_0)}_{MPL \cdot \Delta L} + \epsilon \quad (3)$$

The decomposition implemented in this paper is a discrete approximation of this expansion in *vector space*, where the output image Y should be considered as a location in a multi-dimensional space, captured by its embedding. Equation (4) presents a similar approximation in vector space. $MP_{AI} \cdot \Delta AI$ and $MPL \cdot \Delta L$ retain similar interpretation as in equation (3), while ϵ is the residual of the output shift.

$$\begin{aligned} \vec{Y}_1 - \vec{Y}_0 = & \underbrace{\vec{F}(AI_1, L_0) - \vec{F}(AI_0, L_0)}_{MP_{AI} \cdot \Delta AI} + \underbrace{\vec{F}(AI_0, L_1) - \vec{F}(AI_0, L_0)}_{MPL \cdot \Delta L} \\ & + \underbrace{\vec{F}(AI_1, L_1) - \vec{F}(AI_1, L_0) - \vec{F}(AI_0, L_1) + \vec{F}(AI_0, L_0)}_{\epsilon} \end{aligned} \quad (4)$$

How should we interpret these fractions economically? The magnitude of $MPL \cdot \Delta L$ will inform us how much human adaptation to new AI shifts the output images. The signs of these terms reveal to what extent AI and human input are complements

or substitutes, with human input defined as words in the prompt.

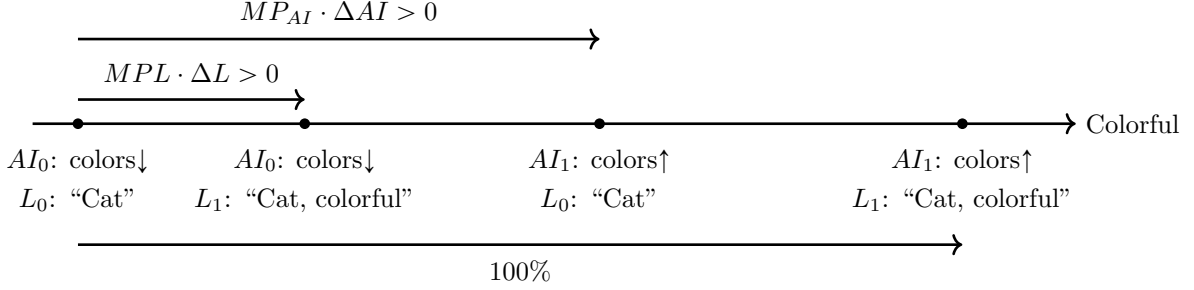


Figure 6: Illustration: AI and Human Input as Complements

Suppose AI and human inputs are complements. Consider the case where the new AI interprets color words more effectively than the old version. In the old version, even if users include color words in the prompt, AI does not understand them, and hence users say *fewer* color words in the old AI. However, as the AI’s understanding of color words improves, users adapt by incorporating more color words into their prompts. Another example may be that in the old version, AI does not recognize certain artist names. As a result, users exclude them from prompts. When the new AI recognizes these names, users include them.

Figure 6 illustrates this scenario. Suppose that in the old AI, the prompt is “Cat”, while in the new AI, the prompt is “Cat, colorful”. When the new prompt is submitted to the old AI, the resulting image shifts toward the new AI’s output. Similarly, when the old prompt is submitted to the new AI, the resulting image also shifts toward the new output. If the total output shift from $Image(AI_0, L_0)$ to $Image(AI_1, L_1)$ is normalized to 100%, then $MP_{AI} \cdot \Delta AI$, the contribution from changing AI alone, is positive and points in the direction of the new image. $MPL \cdot \Delta L$, the contribution from changing the prompt alone, is also positive and points in the same direction. Hence $MP_{AI} \cdot \Delta AI > 0$, $MPL \cdot \Delta L > 0$.

By contrast, suppose AI and human input are substitutes. Consider the case where the old AI generates less realistic images by default, requiring users to specify “Cat, super-realistic” to obtain sufficiently realistic images. In contrast, the new AI automatically generates more realistic images even without explicit realistic speci-

cations in the prompt. Users adapt to this change by including *fewer* realistic words in the prompt.

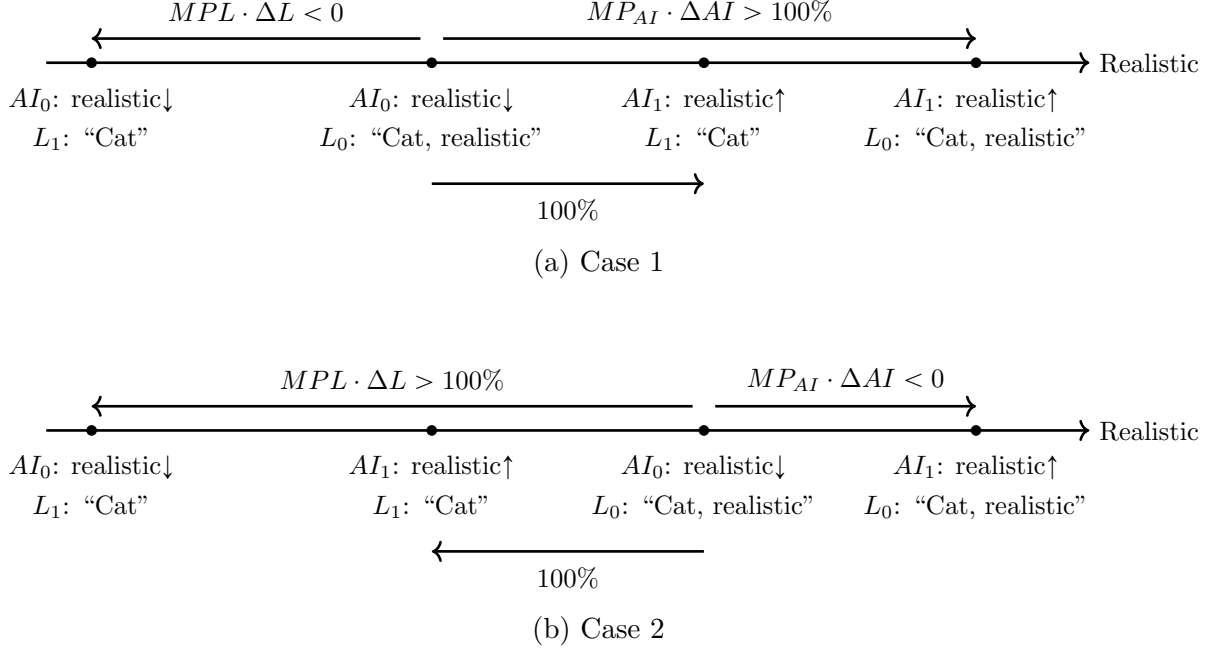


Figure 7: Illustration: AI and Human Input as Substitutes

Figure 7 panel (a) illustrates this scenario. When the new prompt “Cat” is submitted to the old AI, the resulting image is less realistic than the old output. When the old prompt “Cat, realistic” is submitted to the new AI, the resulting image is more realistic than the new output. If the total output shift from $Image(AI_0, L_0)$ to $Image(AI_1, L_1)$ is normalized to 100%, then $MP_{AI} \cdot \Delta AI$, the contribution from changing AI alone, is positive and points toward the new image. However, $MPL \cdot \Delta L$, the contribution from changing the prompt alone, points in the *opposite* direction. Hence, $MP_{AI} \cdot \Delta AI > 0$, $MPL \cdot \Delta L < 0$.

Figure 7 panel (b) illustrates the other possible scenario where the new image is less realistic because users no longer include the word “realistic” in their prompts. In this case, $MP_{AI} \cdot \Delta AI < 0$, $MPL \cdot \Delta L > 0$. More generally, if AI and human inputs are substitutes, $MP_{AI} \Delta AI \cdot MPL \Delta L < 0$.

6.2 Decomposition: Implementation

To implement the decomposition, I randomly select 1000 sessions in V3 and 1000 sessions in V4, collecting the generated images. For each session, I find the final prompt and image by timestamp. I focus on the final prompts and images of the sessions because they represent the final outcome of a creative process. I then submit prompts written for V3 to V4, and prompts written for V4 to V3, collecting the generated images.

To calculate the fractions discussed in the previous section, I match prompts from V3 with prompts from V4 based on similarity. The rationale is to compare similar content: a robot image in V3 should be compared with a robot image in V4, and a landscape in V3 should be compared with a landscape in V4. Moreover, since the decomposition represents a discrete approximation of the first-order expansion of the production function, comparisons should be as local as possible. With 1000 prompts from each version, I construct a 1000×1000 similarity score matrix and apply the Hungarian Algorithm (Munkres, 1957) to obtain a one-to-one mapping. This algorithm has been widely adopted in computer science and engineering for matching strings, such as bus route identifiers and gene sequences. This procedure yields 1,000 matched tuples of the form $\left(Image(AI_{v3}, L_{v3}), Image(AI_{v4}, L_{v4}) \right)$. Some reprompting attempts were unsuccessful because certain words that users originally employed were permitted by Midjourney at the time but were later banned when I reprompted them. These include violent or sexual content and copyrighted elements. In addition, ChatGPT refused to describe some images for sexual, violent, or political reasons. After removing these unsuccessful attempts, I construct 604 tuples of the form $\left(Image(AI_{v3}, L_{v3}), Image(AI_{v4}, L_{v3}), Image(AI_{v3}, L_{v4}), Image(AI_{v4}, L_{v4}) \right)$. Figure D3 in Appendix D displays an example of such a matched tuple.

After constructing matched tuples, I use GPT-4o to generate textual descriptions of each image and convert these descriptions into embeddings using the OpenAI text-embedding-3-small model. The obtained embeddings are projected onto the direction of $Image(AI_{v4}, L_{v4}) - Image(AI_{v3}, L_{v3})$ to obtain $MP_{AI} \cdot \Delta AI$ and $MPL \cdot \Delta L$ for

each tuple. The following section reports the average values of $MP_{AI} \cdot \Delta AI$ and $MPL \cdot \Delta L$ across all matched tuples.

6.2.1 Decomposition: Result

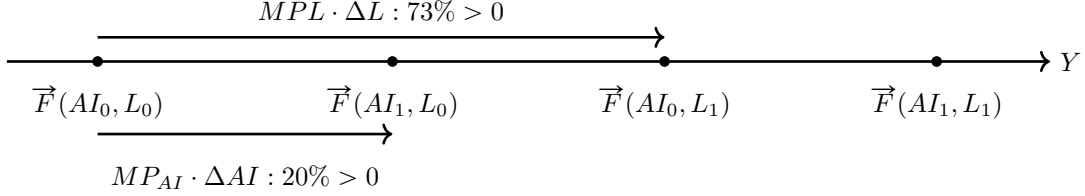


Figure 8: Decomposition Result

$$\epsilon = 100\% - MP_{AI} \cdot \Delta AI - MPL \cdot \Delta L = 7\% \quad (5)$$

Figure 8 and equation (5) present the average values of $MP_{AI} \cdot \Delta AI$, $MPL \cdot \Delta L$, and ϵ . The decomposition attributes 20% of the total output shift to AI changes, 73% to prompt adaptations, and 7% to the residual term. The positive signs of $MP_{AI} \cdot \Delta AI$ and $MPL \cdot \Delta L$ indicate that AI and human input are more like complements than substitutes. The main reason is that both ΔAI and ΔL are positive. It means that the AI's improvement and users' prompt changes reinforce one another. As AI improves at interpreting certain words (e.g., color or photography words), users supply *more* of those words, suggesting complementarity between AI and human input.

Figure 9 displays a heat map of $(MP_{AI} \cdot \Delta AI, MPL \cdot \Delta L)$ across all matched tuples. For 96.5% of tuples, both fractions $MP_{AI} \cdot \Delta AI$ and $MPL \cdot \Delta L$ are positive, indicating that AI and human inputs are complements in the majority of cases.

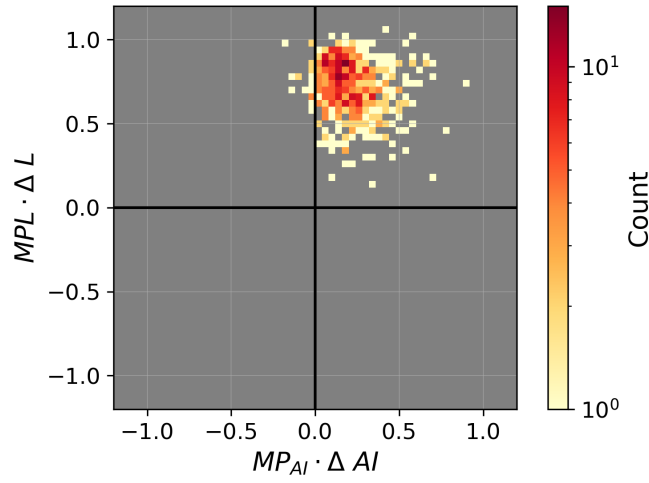


Figure 9: Density Heatmap of AI vs Prompt Contributions

Notes: This figure displays the heat map of $(MP_{AI} \cdot \Delta AI, MPL \cdot \Delta L)$ of the tuples. 96.5% tuples fall within the domain with both fractions being positive.

6.3 Supporting Evidence of Complementarity: Longer Prompts in New AI

$$\# \text{Words}_{ip} = \beta \mathbf{1}(NewAI)_{ip} + \mu_i + \epsilon_{ip} \quad (6)$$

	(1)	(2)	(3)
Dep Var	<i>#Words</i>	<i>#Stop Words</i>	<i>#Non Stop Words</i>
$1(NewAI)$	3.28*** (0.88)	1.08*** (0.16)	2.20*** (0.78)
Mean in Old AI Prompt	19.59	3.07	16.52
User FE	Y	Y	Y
N	49,803	49,803	49,803
R^2	0.31	0.18	0.33

Notes: *** denotes significance at 1 percent, ** at 5 percent, and * at 10 percent. Standard errors are clustered at the user level. This table presents the estimates of regression (6).

Table 1: Number of Words Increases in New AI

To further examine complementarity between AI and human inputs, I estimate regression (6) to test whether users include more words in their prompts in the new version. For each user, I consider the 50 prompts before and the 50 prompts after the V4 release date, constructing a balanced panel around this threshold. The dependent variable is the total number of words in prompt p written by user i .

Table 1 presents the regression results. The number of words per prompt increases by 16.7% (3.28 words) in the new AI.. However, this increase in total word count may not fully reflect the amount of meaningful content in prompts. Column (2) shows that users also include more stop words after the new AI is introduced. This reflects the new AI’s improved understanding of natural language. Hence, users formulate prompts as complete sentences (e.g., “a cat wearing a blue shirt and drinking coffee”) instead of keyword sequences (e.g., “cat, blue shirt, drinking coffee”). Examining total word count alone could therefore only capture that prompts become more like natural language in the new AI.

To better assess whether users provide more informative content, I examine the number of non-stop words in each prompt. Column (3) in Table 1 reports the corresponding regression, using the count of non-stop words as the dependent variable. The results suggest that the number of non-stop words increases significantly in the new AI, implying that users write more informative words in prompts when AI capa-

bility improves. On average, the number of non stop words increases by 2.20 in the new AI. This pattern suggests that users include more meaningful words when the AI becomes more capable, consistent with a complementary relationship between AI performance and human input quality.

Figure 10 plots the estimated coefficients β_k in regression (7). The figure shows a sharp and immediate increase in non-stop words after the new AI release.

$$\# \text{Non-Stop Words}_{ip} = \sum_k \beta_k \mathbb{1}(k^{th} \text{ Prompt})_{ip} + \mu_i + \epsilon_{ip} \quad (7)$$

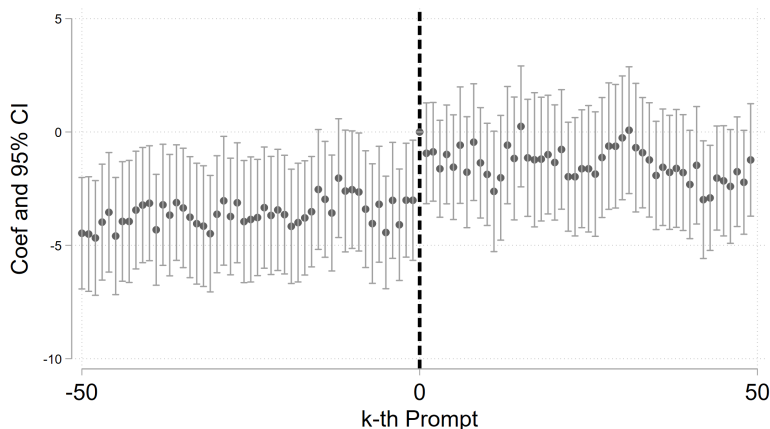


Figure 10: The Number of Non-Stop words Increases in New AI

Notes: This figure plots coefficients before $\mathbb{1}(k^{th} \text{ Prompt})_{ip}$ in regression (7) and the 95% confidence intervals. Standard errors are clustered at the user level.

7 Conclusion

In this paper, I study the role of human adaptation in the creative process when generative AI upgrades. I utilize AI upgrades during the observation period and show that the new AI is significantly more accurate compared to the old AI. When AI capability changes, users also adapt to the new technology and change the words

to include in their prompts. Users include fewer words, such as “realistic” and “detailed” in the new AI, since it automatically produces images with greater realism and detail even without these explicit instructions. Conversely, users include more words about colors and photography in the new AI because it can recognize these words more effectively than before. Thus, when AI is upgraded, both AI capability and human input shift, jointly shifting the output images.

To isolate the effect of AI changes and prompt changes on the final outcome, I decompose the observed output shifts by resubmitting new prompts to the old AI and old prompts to the new AI. The generated images are converted into embeddings with LLM and projected onto the direction of $\overrightarrow{Image_{new}} - \overrightarrow{Image_{old}}$. The decomposition results show that 73% of the output shifts come from changing prompts alone, 20% from changing AI alone, and 7% of residuals, suggesting complementarity between AI and human input. The decomposition highlights the importance of human adaptation for extracting value from new technologies.

More generally, this paper provides a new way to understand the production function without explicitly assuming a functional form. When the two counterfactuals $F(k_{new}, l_{old})$ and $F(k_{old}, l_{new})$ can be observed, a similar decomposition can be implemented to understand whether inputs are complements or substitutes and to quantify the contribution of each input to shifts in output. For example, the same idea could be applied in firms that run experiments to study how technological improvements affect production. In principle, a firm could use experimental variation to measure output under counterfactual combinations of new capital with old labor practices, or old capital with new labor practices, and thereby learn about the channels through which productivity changes appear. However, the empirical setting studied here has an important advantage: the counterfactuals are cheaper to observe. In traditional settings, it would be costly to maintain old machines and ask newly trained workers to operate them in updated ways, or the other way around. By contrast, both counterfactuals $F(k_{new}, l_{old})$ and $F(k_{old}, l_{new})$ can be easily obtained at low cost.

This decomposition highlights the crucial role of human adaptation when technology advances. Simply applying historical prompts to a more advanced AI does not replicate actual outputs, because users continuously refine their language to leverage

the new tool's strengths. In this setting, as AI capability evolves, human judgment remains essential in the creative process. It determines which words are more suitable to include in the new AI.

An implication is that evaluating AI upgrades without accounting for user adaptation underestimates the full potential of the technology. Improvements in model capability may translate into better results when users adjust their inputs.

Appendix

A Data Appendix

A.1 Data Collection

Data collection started in February 2, 2025 and ended in February 8, 2025. I collected messages from channels called “general 1”, “general 2”, ..., “general 20”.

A.2 Clean Errors And Link Type 19 Messages to Type 0 Messages

The data cleaning process follows the following steps.

1. Read all messages.
2. Delete messages whose type is not 0 or 19. Type 0 messages are those prompts submitted by users. Type 19 messages are the result of clicked buttons. I also delete those that are not sent by the Midjourney Bot and those with errors.
3. For each message of type 19, identify the *message_reference* to determine which message the user interacted with to trigger the current message. By doing so, I can trace backward until a type 0 message is found. For a given type 0 message, if there is more than one user involved along the path, meaning that more than one user clicks on the buttons, I remove this type 0 message and all the subsequent messages.
4. Break “content” in each message into text inputs, parameters, image inputs, and mode.
5. For each type 0 message, if at least one image input comes from the generated output of another user, remove the path (keep the original user’s path).

A.3 Clustering Prompts Into Sessions

This section describes the details of clustering prompts into sessions. Table A1 presents the distribution of session time spans.

1. For text inputs in each type 0 message, remove stop words and “magic words”.
 - Stop words are words like “a”, “an”, “the”, “here”. This is a terminology usually employed by computer scientists who study natural language processing. I am using the list from the “nltk.corpus” package in Python.
2. For a given user, collect all type 0 messages and find N cleaned text inputs. Calculate a $N \times N$ Jaccard similarity matrix. The Jaccard similarity score is between 0 and 1, with a larger number meaning texts being more similar.
 - For each type 0 message, if at least one of the image inputs comes from the generated output of the **same user**, assign a similarity score of one to guarantee that they are in the same session.
3. For prompt pairs in which at least one of them does not have text input, look at the image inputs. If at least one of the image inputs is the same, assign a similarity score of one to guarantee that they are in the same session.
4. Use DBSCAN to cluster type 0 messages within a user. Assign the threshold to be 0.2.
5. For each session, find the first action by timestamp and remove it if it is within the last 24 hours of the data collection period. This is to remove those sessions that have just started, but I did not collect the complete path due to a fixed data collection start time.

Session Time Span	%
0–1 h	90.62
1–6 h	1.30
6–12 h	0.37
12–24 h	0.55
24–48 h	0.47
48–72 h	0.25
3–7 days	0.56
7–30 days	1.11
> 30 days	4.77

Table A1: Distribution of Session Time Span

A.4 Word List For Classifications

Below are some examples of the word classifications. The complete list will be available in the online appendix.

1. *detailed words*: detail, detailed, high-detailed, ultra-detailed, hyper-detailed, extremely-detailed
2. *realistic words*: realistic, hyperrealistic, super realistic, photorealistic, photorealism
3. *rendering engines*: rendering engine, unreal engine, Cinema4D, C4D, Arnold Render, Octane Render
4. *colors*: vibrant colors, black, white, emerald, olive-green, vivid-maroon, cyan, neon colors, light yellow
5. *stop words*: in, on, just, me, as, is, are, am, there, here, had, while
6. *photography words*: fisheye lens effect, 135mm lens, white balance, shutter speed, stop motion

7. *lighting*: tyndall effect, dim lighting, glowing radioactivity, studio light, dj lighting
8. *style*: post-impressionism, neo-pop art, street art, futurism, cubist, dadaism

A.5 Clustering Sessions into Topics

I classify sessions into topics using a popular language-processing Python package called BERTopic. It is widely used in classifying documents into topics using clustering.

For prompts in a single session, I append the text inputs together. And then I remove the color words and stop words to improve clustering accuracy. I then cluster sessions into 100 topics using BERTopic, with the K-Means clustering method. Below are some examples of topics, with corresponding keywords, and some sampled prompts in each topic.

1. Landscapes

Keywords: mountains, lake, sky, landscape, desert, river, sunset, water

Sampled Prompts:

- “new Mexico landscape, clear blue sky, rocky mountains, rainclouds, dramatic, oil painting, detailed, Albert bierstadt style”
- “dramatic cloud formation coming from behind a mountain, clear blue sky, midday, bright, southwest, colorful, awe-inspiring, detailed, oil painting”

2. Cute Animals

Keywords: cat, dog, cute, kitten, adorable, puppy, fur, bunny

Sampled Prompts:

- “super cute black french bulldog + big expressive eyes + Pixar + ghibli + extremely detailed + detailed reflections + bold highlights + photo-realistic + hdr + octane render + 8K + soft light + twinkling fire + smoke”

- “white cat, roller skating, playing keytar, wearing sunglasses, playing keytar”

3. Architectures and Room Designs

Keywords: room, interior, house, modern, design, walls, furniture, architecture, wall

Sampled Prompts:

- “studio with architects designing models of future cities”
- “luxurious comfortable biopunk lounge inside a huge tropical greenhouse, volumetric lighting, hyperdetailed, 8k octane render”

4. Portraits

Keywords: hair, eyes, face, woman, portrait

Sampled Prompts:

- “tall 50 year old female police officer with Curly blonde hair and wearing black framed reading glasses, realistic, Norman Rockwell style, 10k”
- “long red hair, beautiful woman in green dress”

B Additional Tables and Figures

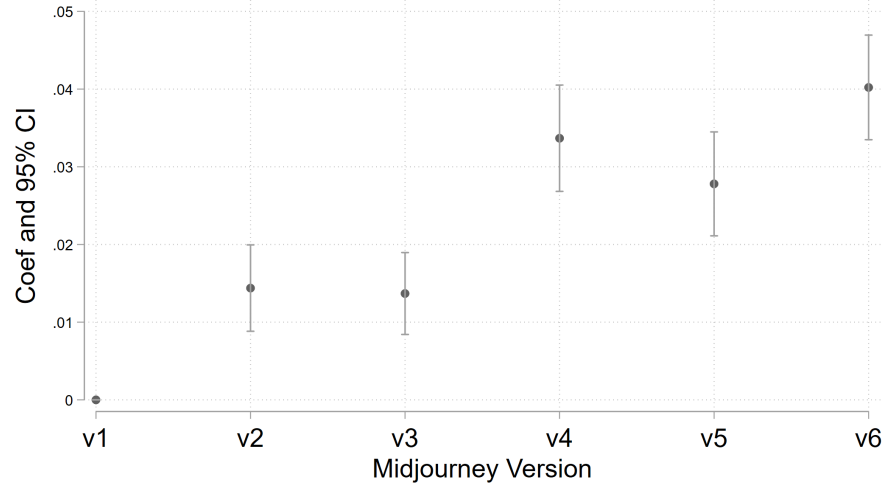


Figure B1: Later Versions Give More Accurate Results

Notes: This figure plots coefficients before $1(\text{Midjourney version} = v)$ in regression (1) and the 95% confidence intervals. Standard errors are clustered at the prompt level. This figure uses embeddings all-MiniLM-L6-v2.

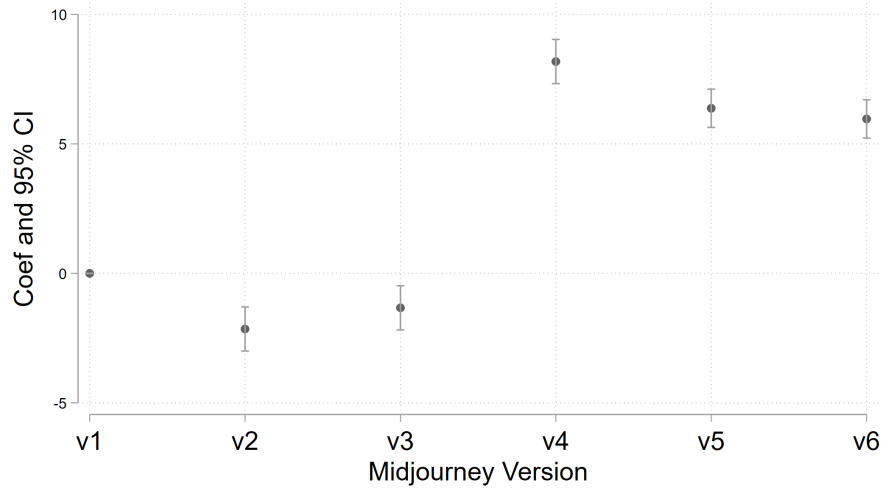


Figure B2: Later Midjourney Versions Give More Detailed Images

$$\%Upscale_{is} = \sum_k \beta_k \mathbb{1}(k^{th} \text{ Session})_{is} + \mu_i + \epsilon_{is} \quad (B1)$$

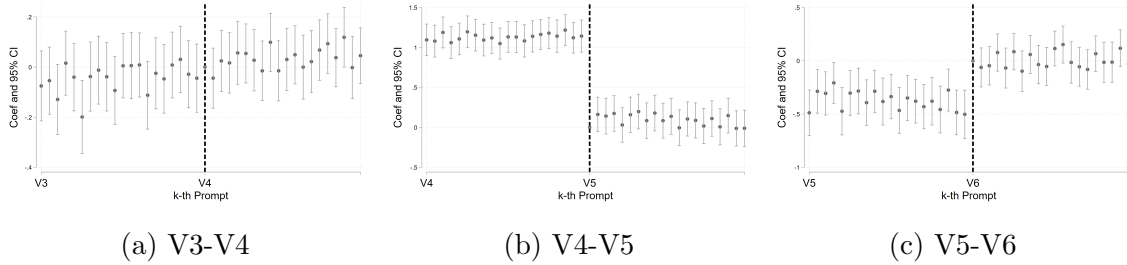


Figure B3: Upscale Ratio Changes across Versions

Notes: This figure plots coefficients before $\mathbb{1}(k^{th} \text{ Session})$ in regression (B1) and the 95% confidence intervals. Note that in V3, default resolution before upscaling is 256×256 pixels, V4 is 512×512 pixels, V5 is 1024×1024 pixels, V6 is 1024×1024 pixels. Hence, the incentives for upscaling could decrease between V3 and V4, V4 and V5. The only versions that are comparable are V5 and V6.

	Mean	Median	99th Percentile	Min	Max
#sessions per user	191	114	1,408	7	4,120
#prompts per session	12	1	18	1	45,720
#words per prompt	28	18	195	0	378
#parameters per prompt	1.60	1	5	0	10
#image inputs per prompt	0.21	0	3	0	57
N_{user}	500				
$N_{session}$	95,545				
N_{prompt}	1,160,773				

Table B1: Summary Statistics

Time	Text	Color	Style	...
0	Mountain, Black and White	Black and White	-	
1	Mountain, Black and White, Realistic	Black and White	Realistic	
2	Mountain, Neon Color, Abstract	Neon Color	Abstract	
3	Mountain, Neon Color, Abstract	Neon Color	Abstract	

Table B2: Illustration of Splitting Texts

C Screenshots About User Composition

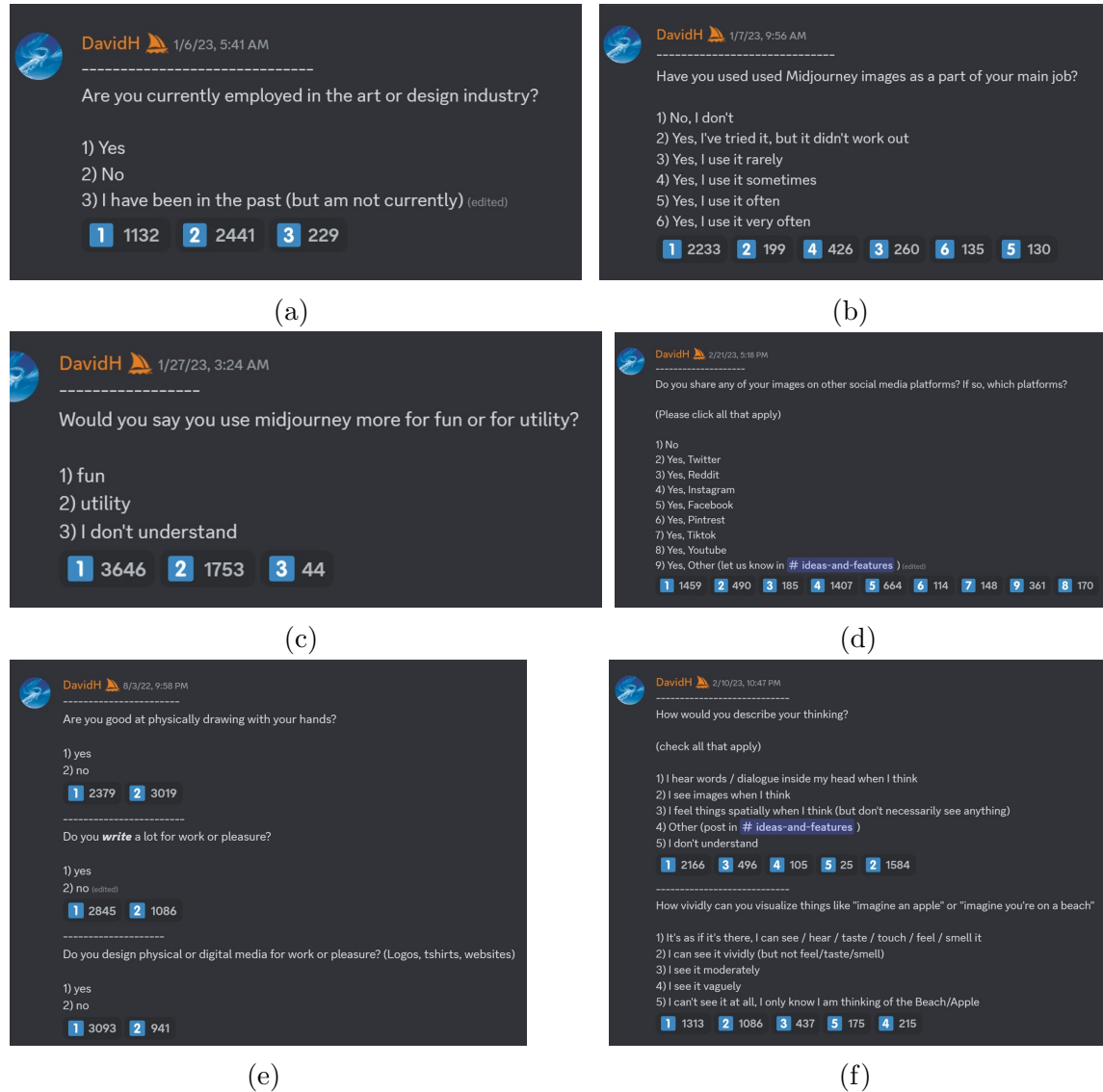
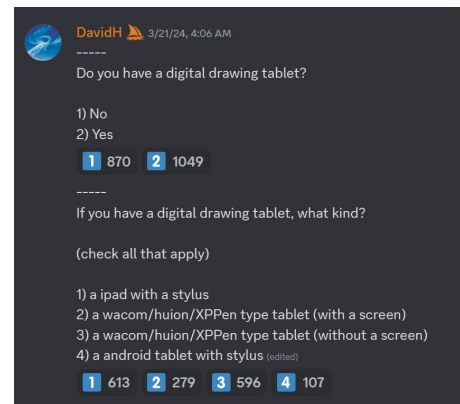


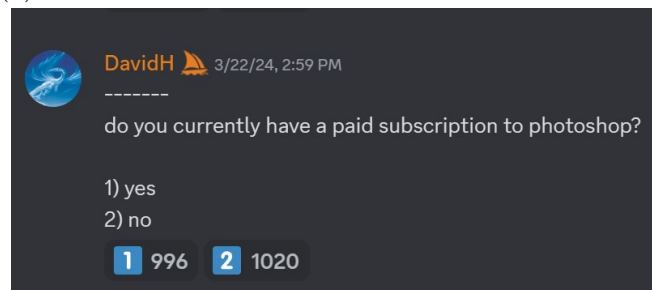
Figure C1: "Mini Poll" Screenshots 1



(a)



(b)



(c)

Figure C2: “Mini Poll” Screenshots 2

midjourney.com Compare

Audience Interests

Oct, 2023 - Dec, 2024 (Custom) United States

	Domain (9,561)	Industry	Global Rank	Relevance Score	Cross-Visitation	PoP change	AdSense
1	app.runwayml.com	Games > Video Games Consoles and Acco...	-	100	3.22%	-	-
2	app.leonardo.ai	Computers Electronics and Technology > C...	-	98	3.15%	-	-
3	elevenlabs.io	Computers Electronics and Technology > P...	#3,890	90	2.95%	-	-
4	runwayml.com	Computers Electronics and Technology > G...	#6,988	89	2.25%	-	-
5	printify.com	Business and Consumer Services > Busine...	#5,806	86	2.74%	-	-
6	labs.openai.com	Games > Video Games Consoles and Acco...	-	80	2.34%	-	-
7	ideogram.ai	Computers Electronics and Technology > P...	#4,336	79	2.11%	-	-
8	freemik.com	Arts and Entertainment > Arts and Entertai...	#225	76	2.95%	-	-
9	openai.com	Computers Electronics and Technology > C...	#88	75	7.61%	-	-
10	huggingface.co	Computers Electronics and Technology > C...	#2,379	74	2.67%	-	-
11	leonardo.ai	Computers Electronics and Technology > C...	#4,269	73	1.73%	-	-
12	canva.com	Computers Electronics and Technology > G...	#39	72	15.28%	-	-
13	claude.ai	Computers Electronics and Technology > P...	#797	71	4.66%	-	-
14	platform.openai.com	Computers Electronics and Technology > P...	-	67	3.20%	-	-
15	lumalabs.ai	Computers Electronics and Technology > P...	#10,244	66	1.60%	-	-
16	checkout.stripe.com	Computers Electronics and Technology > C...	-	66	7.18%	-	-
17	civitai.com	Computers Electronics and Technology > S...	#2,340	65	1.93%	-	-
18	behance.net	Arts and Entertainment > Visual Arts and D...	#1,314	65	2.07%	-	-
19	openart.ai	Computers Electronics and Technology > C...	#9,849	64	1.61%	-	-
20	freem.com	Computers Electronics and Technology	#894	63	3.27%	-	-
21	auth0.openai.com	Computers Electronics and Technology > P...	-	63	6.00%	-	-
22	freemfy.adobe.com	Computers Electronics and Technology > G...	-	62	1.46%	-	-

(a)

docs.midjourney.com Compare

Audience Interests

Oct, 2023 - Dec, 2024 (Custom) United States

	Domain (2,129)	Industry	Global Rank	Relevance Score	Cross-Visitation	PoP change	AdSense
1	midlibrary.io	Computers Electronics and Technology > S...	#202,470	100	4.84%	-	-
2	app.runwayml.com	Games > Video Games Consoles and Acco...	-	97	5.87%	-	-
3	checkout.stripe.com	Computers Electronics and Technology > C...	-	95	13.37%	-	-
4	runwayml.com	Computers Electronics and Technology > G...	#6,988	94	5.06%	-	-
5	app.leonardo.ai	Computers Electronics and Technology > C...	-	89	5.43%	-	-
6	openart.ai	Computers Electronics and Technology > C...	#9,849	85	4.81%	-	-
7	bootcamp.udesign.cc	Computers Electronics and Technology > C...	-	84	3.46%	-	-
8	prompthero.com	Computers Electronics and Technology > G...	#43,763	83	3.93%	-	-
9	ideogram.ai	Computers Electronics and Technology > P...	#4,336	81	4.86%	-	-
10	stability.ai	Computers Electronics and Technology > P...	#51,585	79	3.83%	-	-
11	leonardo.ai	Computers Electronics and Technology > C...	#4,269	76	3.42%	-	-
12	printify.com	Business and Consumer Services > Busine...	#5,806	76	5.68%	-	-
13	promptbase.com	Computers Electronics and Technology > C...	#119,644	75	3.36%	-	-
14	elevenlabs.io	Computers Electronics and Technology > P...	#3,890	72	5.14%	-	-
15	help.openai.com	Computers Electronics and Technology > P...	-	70	4.27%	-	-
16	labs.openai.com	Games > Video Games Consoles and Acco...	-	70	3.20%	-	-
17	civitai.com	Computers Electronics and Technology > S...	#2,340	69	5.02%	-	-
18	pika.art	Computers Electronics and Technology > G...	#20,537	67	3.74%	-	-
19	zapier.com	Computers Electronics and Technology > P...	#6,214	67	5.24%	-	-
20	creator.nightcafe.studio	Computers Electronics and Technology > G...	-	67	3.68%	-	-
21	support.discord.com	Games > Video Games Consoles and Acco...	-	66	7.54%	-	-
22	craiyon.com	Games > Video Games Consoles and Acco...	#22,589	65	2.85%	-	-

(b)

Figure C3: SimilarWeb Screenshots

D Image Illustrations

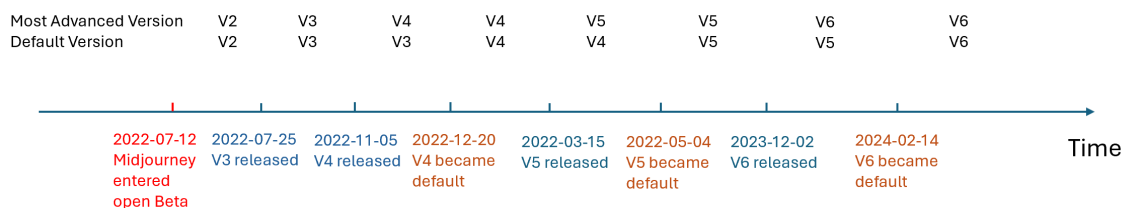
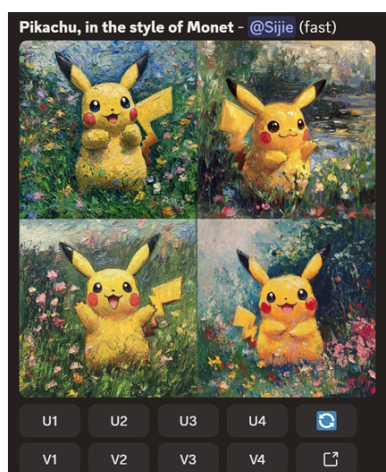


Figure D1: Timeline of Version Updates

If I type “Pikachu, in the style of Monet”
in the chat box:



Pikachu, captain hat, style of Mucha **Action 1**
11/24/24 3:33pm



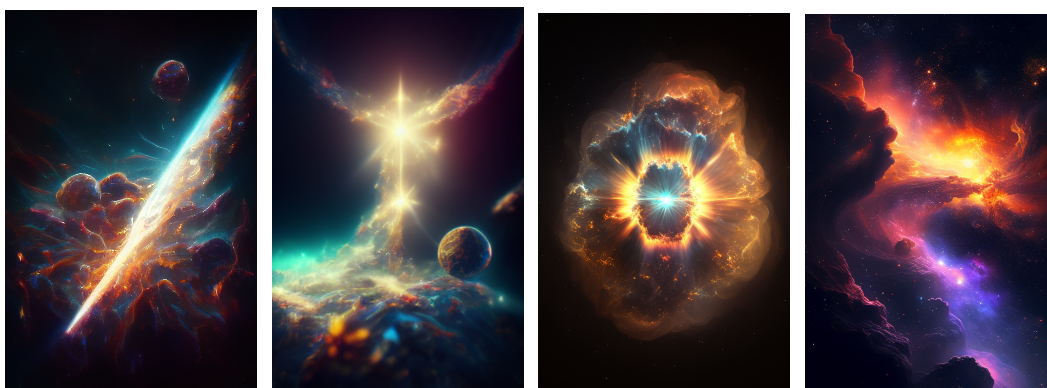
Make variation 3 **Action 2**
11/24/24 3:34pm



Upscale 1 **Action 3**
11/24/24 3:35pm



Figure D2: An Example of Using Midjourney



(a) **AI3 Prompt3:** “supernova,8k” (b) **AI3 Prompt4** (c) **AI4 Prompt3** (d) **AI4 Prompt4:**
“ethereal celestial
cosmic space, 8k,
hyper detail, hdr,
cinematic, high reso-
lution”

Figure D3: An Example of Matched Tuple

References

- Acemoglu, D., Autor, D., Hazell, J., & Restrepo, P. (2022). Artificial Intelligence and Jobs: Evidence from Online Vacancies. *Journal of Labor Economics*, 40(S1), S293–S340. doi:10.1086/718327
- Acemoglu, D., & Restrepo, P. (2020). Robots and Jobs: Evidence from US Labor Markets. *Journal of Political Economy*, 128(6), 2188–2244. doi:10.1086/705716
- Akerberg, D. A., Caves, K., & Frazer, G. (2015). Identification Properties of Recent Production Function Estimators. *Econometrica*, 83(6), 2411–2451. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.3982/ECTA13408>. doi:10.3982/ECTA13408
- Agrawal, A., Gans, J. S., & Goldfarb, A. (2019). Artificial Intelligence: The Ambiguous Labor Market Impact of Automating Prediction. *Journal of Economic Perspectives*, 33(2), 31–50. doi:10.1257/jep.33.2.31
- Angelova, V., Dobbie, W. S., & Yang, C. (2023, October). *Algorithmic Recommendations and Human Discretion* (tech. rep. No. w31747). National Bureau of Economic Research. doi:10.3386/w31747
- Babina, T., Fedyk, A., He, A., & Hodson, J. (2024). Artificial intelligence, firm growth, and product innovation. *Journal of Financial Economics*, 151, 103745. doi:10.1016/j.jfineco.2023.103745
- Brynjolfsson, E., Li, D., & Raymond, L. (2025). Generative AI at Work*. *The Quarterly Journal of Economics*, 140(2), 889–942. doi:10.1093/qje/qjae044
- Chatterji, A., Cunningham, T., Deming, D., Hitzig, Z., Ong, C., Shan, C., & Wadman, K. (2025, September). How People Use ChatGPT. SSRN Scholarly Paper. doi:10.2139/ssrn.5487080
- Choi, J. H., & Schwarcz, D. (2024). AI Assistance in Legal Analysis: An Empirical Study. *Journal of Legal Education*. Retrieved October 19, 2025, from https://librarysearch.library.utoronto.ca/nde/fulldisplay/cdi_scopus_primary_2_s2_0_105004031311/01UTORONTO_INST:UTORONTO_NDE/en
- Dell’Acqua, F., McFowland, E., Mollick, E. R., Lifshitz-Assaf, H., Kellogg, K., Rajendran, S., . . . Lakhani, K. R. (2023). Navigating the Jagged Technological Fron-

- tier: Field Experimental Evidence of the Effects of AI on Knowledge Worker Productivity and Quality. *SSRN Electronic Journal*. doi:10.2139/ssrn.4573321
- Handa, K., Tamkin, A., McCain, M., Huang, S., Durmus, E., Heck, S., . . . Ganguli, D. (2025, February). Which Economic Tasks are Performed with AI? Evidence from Millions of Claude Conversations. arXiv:2503.04761 [cs]. doi:10.48550/arXiv.2503.04761
- Hsieh, C.-T., & Klenow, P. J. (2009). Misallocation and Manufacturing TFP in China and India*. *The Quarterly Journal of Economics*, 124(4), 1403–1448. doi:10.1162/qjec.2009.124.4.1403
- Jahani, E., Manning, B. S., Zhang, J., TuYe, H.-Y., Alsobay, M., Nicolaides, C., . . . Holtz, D. (2025, April). Prompt Adaptation as a Dynamic Complement in Generative AI Systems. arXiv:2407.14333 [cs]. doi:10.48550/arXiv.2407.14333
- Kanazawa, K., Kawaguchi, D., Shigeoka, H., & Watanabe, Y. (2025). AI, Skill, and Productivity: The Case of Taxi Drivers. *Management Science*. doi:10.1287/mnsc.2023.01631
- Levinsohn, J., & Petrin, A. (2003). Estimating Production Functions Using Inputs to Control for Unobservables. *The Review of Economic Studies*, 70(2), 317–341. doi:10.1111/1467-937X.00246
- Lin, S. (2026, August). Human Adaptation in Creative Process with Generative AI. SSRN Scholarly Paper. Rochester, NY: Social Science Research Network. Retrieved August 17, 2026, from <https://papers.ssrn.com/abstract=7299800>
- Munkres, J. (1957). Algorithms for the Assignment and Transportation Problems. *Journal of the Society for Industrial and Applied Mathematics*, 5(1), 32–38.
- Noy, S., & Zhang, W. (2023). Experimental evidence on the productivity effects of generative artificial intelligence. *Science*, 381(6654), 187–192. doi:10.1126/science.adh2586
- Olley, G. S., & Pakes, A. (1996). The Dynamics of Productivity in the Telecommunications Equipment Industry. *Econometrica*, 64(6), 1263–1297. doi:10.2307/2171831

- Peng, S., Kalliamvakou, E., Cihon, P., & Demirer, M. (2023, February). The Impact of AI on Developer Productivity: Evidence from GitHub Copilot. arXiv:2302.06590 [cs]. doi:10.48550/arXiv.2302.06590
- Vaccaro, M., Almaatouq, A., & Malone, T. (2024). When combinations of humans and AI are useful: A systematic review and meta-analysis. *Nature Human Behaviour*, 8(12), 2293–2303. doi:10.1038/s41562-024-02024-1
- Wooldridge, J. M. (2009). On estimating firm-level production functions using proxy variables to control for unobservables. *Economics Letters*, 104(3), 112–114. doi:10.1016/j.econlet.2009.04.026
- Zhou, E. B., & Lee, D. (2023, October). Generative AI, Human Creativity, and Art. SSRN Scholarly Paper. doi:10.2139/ssrn.4594824
- Zhou, E. B., Lee, D., & Gu, B. (2025). Who expands the human creative frontier with generative AI: Hive minds or masterminds? *Science Advances*, 11(36), eadu5800. doi:10.1126/sciadv.adu5800